



**Ref:** SMTTP/ELECT MAINT/2023-24/LMI/528517

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**Subject:** Revision of LMI/OGN/OPS/ELEC/014, Shaft voltage and currents in large turbo generators

Revision of LMI/OGN/OPS/ELEC/014, Shaft voltage and currents in large turbo generators

LMIs are reviewed periodically for further improvement and incorporation of Recommendation by internal or external audit committee and COS.

OGN/OPS/ELEC/014 has been revised by COS in April 2022. No changes envisaged by COS.  
LMI is reviewed.

Records of revision:

- a) Frequency of shaft voltage measurement changed from monthly to quarterly.
- b) Generator voltage and power factor reading added in checklist annexure-I.

Put up for kind approval.

Attachments:

1. LMI/OGN/OPS/ELEC/014, Issue-3, Rev 3, Oct 2023
2. OGN/OPS/ELEC/014, Rev 1, May 2020

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**Note No:2**

reviewed and necessary changes done.

Revision number updated.

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LMI Reviewed.

May be approved

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LMI reviewed and revised Oct23, check list enclosed.

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**Note No:7**

Revised LMI may please be accorded approval.

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**NTPC LIMITED  
SIMHADRI SUPER THERMAL POWER STATION**

**LOCATION MANAGEMENT INSTRUCTION  
LMI/OGN/ELEC/014  
ISSUE NO.: 3, REVISION 3, DATE: OCT 2023**

**Shaft Voltages and Current in large  
Turbo Generators**

APPROVED FOR

IMPLEMENTATION \_ \_ \_ \_ \_

By GROUP GENERAL MANAGER (SMTTP)

DATE \_ \_ \_ \_ \_

**NTPC LIMITED.**

**SIMHADRI SUPER THERMAL POWER STATION**

**OPERATION AND MAINTENANCE DEPARTMENT**

|                    |                                                             |              |                |
|--------------------|-------------------------------------------------------------|--------------|----------------|
| <b>LMI(Name) :</b> | <b>Shaft Voltages and Current in large Turbo Generators</b> |              |                |
| <b>LMI No.</b>     | LMI/OGN/ELEC/014                                            |              |                |
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|             | <b>DESIGNATION</b>     | <b>SIGNATURE &amp; NAME</b> |
|-------------|------------------------|-----------------------------|
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## Shaft voltages and currents in large turbo generator

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## **1.0 INTRODUCTION:**

The flow of current, which can cause damages to the bearings and governor gear, from the turbine generator shaft to ground, can occur due to following reasons.

### **1.1 Cross Magnetization of the Turbine:**

The residual magnetization of the turbine shaft, while rotating can develop a voltage along its length. When the turbine shaft gets two low resistance points to the earth potential and the shaft voltage gets short circuited through the turbine housing, current may flow in such a direction, as to induce a shaft voltage in the same direction, as the original voltage.

This is a self excitation condition which can cause very high currents and cause serious damage to the turbine. There have been a few such incidences of damage reported elsewhere.

### **1.2 Generator Magnetic Asymmetry:**

Due to magnetic asymmetry a small voltage will exist across the length of the generator shaft. To prevent circulation of current through the bearings, driven by this voltage either one or both bearings (or the bearing pedestal) are insulated from ground. All the oil pipe line, oil seals, which come in contact with generator shaft are also insulated. In the case of machines with exciter coupled, the coupling is also normally insulated. It is difficult to measure the resistance of this insulation as the shaft is effectively earthed at a number of locations through the turbine.

### **1.3 Static charge developed on the Turbine blades:**

Due to the frictional flow of steam and droplets of water being thrown over the turbine blades in the last stages a static voltage is produced, which can build up

to high values and result in discharges to earth potential, through the bearings, puncturing the oil film.

For KWU turbine at SIMHADRI, the grounding is provided between LP Turbine and Generator.

To avoid damages due to the above reasons to the bearing and the governing system, the shaft voltage measurement is done.

## **2.0 SCOPE:**

This LMI covers the Unit # 1, 2, 3 & 4, 500MW water cooled generator shaft voltages & current.

## **3.0 REFERENCE DOCUMENTS:**

a) Operation guidance note - OGN/OPS/ELEC/014.

## **4.0 SUPERSEDED DOCUMENTS:** LMI/OGN/ELEC/014, ISSUE-3, REV-2, MAY2022

## **5.0 SPECIFIC INSTRUCTIONS TO BE FOLLOWED:**

The shaft grounding brushes which operate in a hostile environment and do not carry current normally, are likely to become poor conductor in the course of time. Periodic checking (Ref 8.0), cleaning of the area and inspection of brushes has to be done to avoid this situation.

The shaft grounding shall not be made at more than one location.

## **6.0 SHAFT GROUNDING PROCEDURE:**

The bearings at either end of the Generator together with seals, oil pipe lines are insulated. Generator shaft grounding is achieved through 2 brush holders with 4 carbon brushes at bearing no 5 . The Brush holders and earth leads shall be well insulated to avoid alternative ground paths.

## **7.0 METHODOLOGY OF IMPLEMENTATION:**

To ensure the rotor grounding arrangement at the turbine end, the shaft voltage should also be measured at exciter end w.r.t turbine end. The shaft voltage measured in this way should then be compared with the voltage measured at exciter end w.r.t earth.

The monitoring of shaft voltages is the simplest way of determining whether the bearing insulation is healthy or not. The shaft voltage measurement will be done when the machine is at full load. Similar machines in a station could also form a basis for comparison. Low shaft voltage in any machine should be investigated during an outage to confirm whether the same is inherent for the machine.

The contact of the earthing brushes will be periodically checked (Ref 8.0).

The carbon brushes used in this application are special silver/copper loaded graphite and care should be taken to ensure correct replacement.

## **8.0 FREQUENCY OF CHECKING AND MEASUREMENTS:**

Shaft voltage monitoring needs to be carried out once in every month at full load with reactive power, frequency, Generator Rotor field current and excitation field current (annexure-1).

The shaft voltage measured at exciter end w.r.t turbine end needs to be carried out quarterly. (annexure-1 Sl.No. 1A).

When shaft voltage falls below 75% of the minimum reference value for the machine, the insulation should be checked at the earliest opportunity by monitoring the carbon brush healthiness and checking of the ground connection.

## **9.0 THE RESPONSIBILITY CENTRES FOR IMPLEMENTATION:**

Basic responsibility lies with respective Area in-charge.

## **10.0 THE EXCEPTION REPORTING & DISPENSATION OF DEVIATIONS:**

Any deviation shall be reported to HOD (EM).

## **11.0 RECORD KEEPING:**

Records will be kept in respective LMI file generator group- Electrical Maintenance

## **12.0 REVIEW:**

Documents will be reviewed once in two years by HOP.

## **13.0 RECORDS OF REVISION**

- a) Frequency of shaft voltage measurement changed from monthly to quarterly.
- b) Generator voltage and power factor reading added in checklist annexure-I.

## **14.0 Annexure-I**

### **GENERATOR SHAFT VOLTAGES**

**Date:**

| S.No. | DESCRIPTION                                   | UNIT- | UNIT- |
|-------|-----------------------------------------------|-------|-------|
| 1     | Shaft to earth voltage (EE Shaft to earth)    |       |       |
| 1A    | Shaft to earth voltage (EE Shaft to TE Shaft) |       |       |
| 2     | MW                                            |       |       |
| 3     | MVAR                                          |       |       |
| 4     | Frequency                                     |       |       |
| 5     | Generator rotor field current                 |       |       |
| 6     | Excitation field current                      |       |       |
| 7     | Generator voltage                             |       |       |
| 8     | Power factor                                  |       |       |